Simulation of a Chestband ECG and Breathing Frequency Monitoring System

A Preprint

Anushka Mukherjee
MSc. Artificial Intelligence for Industrial Applications
OTH Amberg-Weiden
a.mukherjee@oth-aw.de

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Abstract

This report describes a simulated chestband device that measures ECG (heart activity) and breathing rate together, and shows the results in a live desktop dashboard. The system is written in Python. It generates a synthetic raw chest-expansion signal and a synthetic ECG signal, estimates the breathing rate from the raw signal using peak detection, classifies the breathing rate as Low, Normal, or High, and displays the result in real time. Over a simulated 30-day period (43,200 minutes), the average breathing rate was 16.20 breaths/min, with 957 High readings, 16 Low readings, and 17 sustained High episodes of at least 10 minutes. The report explains the theory behind the chosen thresholds and detection method, the system design, a waveform bug that was found and fixed, and the main strengths and limitations of the project.

Keywords Chestband · ECG Simulation · Respiratory Monitoring · Impedance Pneumography · Wearable Health Monitoring · Python

1 Introduction

Wearable chest straps that measure both ECG and breathing rate are an established type of health device, used in medicine and sports to track heart rate, ECG shape, and breathing frequency from one elastic band (Avila and Sampogna, 2011; Zheng et al., 2017). This project simulates the software side of such a device: a data pipeline that creates a realistic raw signal, estimates breathing rate and heart rate from that signal (the same way a real chestband would), classifies the breathing rate, and shows the result in a live application.

No real patient data is used. Every signal is generated by the program, so the true breathing rate is always known and can be compared with the estimated one. This makes it possible to check how well the detection method works, not just to display pre-set numbers.

The signal processing chain selected for this report is the **breathing-rate estimation chain**. As shown in Fig. 1, this chain begins with a raw simulated chest-expansion signal, which is passed through a peak detector to produce a breathing-rate estimate; that estimate is then classified as Low, Normal, or High and shown on the live readout. This is the chain explained and analysed in detail in this report. The simulated ECG and its R-peak heart-rate estimate are also part of the running simulation and are mentioned for context, but they are not the chain this report focuses on.

2 Theory

A healthy adult at rest normally breathes between 12 and 20 times per minute; rates outside this range can be an early sign of a health problem and are often under-monitored in practice (Cretikos et al., 2008). This project uses exactly this range for its three classes: Low ($<$ 12/min), Normal (12–20/min), High ($>$ 20/min).

Real chestbands often measure breathing using impedance pneumography: a small current is passed through the chest, and its resistance changes slightly with every breath. This produces a respiration waveform, from which the breathing rate is counted (Hieronymi et al., 2011). A single trigger point is not reliable, because small artifacts or even the heartbeat itself can be miscounted as a breath. Using both a minimum peak height and a minimum time between peaks is safer (Hieronymi et al., 2011). This project's peak detector uses exactly this two-condition idea.

For the ECG side, the R peak of the QRS complex is the clearest and most common feature used to estimate heart rate (Kramme, 2011). The classic method for finding R peaks in a noisy ECG signal is the Pan–Tompkins algorithm, which filters, squares, and applies changing thresholds to detect each heartbeat reliably (Pan and Tompkins, 1985). This project's heart-rate estimator is a simpler version of the same idea: it looks for peaks above a fixed height with a minimum spacing, then converts the average time between peaks into beats per minute.

A close real-world example of this whole idea is the "Smart Chest Strap" described by Zheng et al. (2017): a chest band that measures ECG, a breathing signal, and movement, and sends the result to a phone. The Equivital system in Avila and Sampogna (2011) does something similar. This project follows the same idea (ECG + breathing rate + live readout), but fully simulated for a coursework setting.

3 Method

The project has three Python scripts and one notebook, all sharing one `data/` folder (Fig. 1):

- `simulate.py` creates 30 days of data (43,200 rows). For every minute it makes a heart rate, a hidden "true" breathing rate, and a raw chest-expansion signal built from that true rate. The breathing rate saved in the file is not the hidden value – it is the value *estimated* from the raw signal by the peak detector, so the estimation step is really tested.
- `analyze.py` reads the CSV and produces a daily summary, a text report, and two charts: the 30-day trend and the status distribution.
- `monitor.py` is a live dashboard that replays the data minute by minute: a scrolling ECG wave with an estimated heart rate, a scrolling breathing wave, and the 30-day trend, with an alert for sustained High episodes.

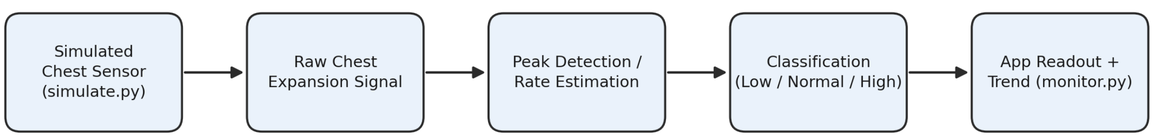

Figure 1: Data flow through the simulation, estimation, classification, and readout stages.

Breathing rate estimation. A 60-second raw signal is generated at 5 Hz for every minute. A peak is only counted if it is tall enough *and* at least 1.5 s after the last accepted peak. The average time between peaks gives the breathing rate. This mirrors the dual-threshold idea from Section 2 (Hieronymi et al., 2011).

A waveform bug that was found and fixed. During testing, the live breathing wave looked wrong: the two ends stayed still while only the middle moved, like a small accordion. The cause was that the wave was redrawn from $t = 0$ every frame (about every 50 ms), so the phase reset each time instead of continuing smoothly. The fix uses the same idea already used correctly for the ECG wave: one running phase value is increased a little every frame, and only one new point is added to a fixed buffer, instead of redrawing the whole wave. This gives a smooth, continuous scroll even when the breathing rate changes.

4 Results

Table 1 shows the full 30-day (43,200-minute) simulation.

Most readings (97.7%) fall in the Normal range, which matches how the hidden true rate was designed: close to 16 breaths/min except during a few scheduled High episodes and rare night-time dips. Because Low and High readings are such a small share of all readings, a normal bar chart makes them look almost invisible. Figure 3 therefore uses a log-scale y-axis with the exact count printed above each bar, so all three classes stay readable. The 17 sustained High

Table 1: Summary statistics for the 30-day simulated dataset

Metric	Value
Total readings	43,200
Mean breathing rate	16.20 breaths/min
Min / max breathing rate	10.56 / 27.18 breaths/min
Normal / High / Low readings	42,227 / 957 / 16
Sustained High episodes (≥10 min)	17

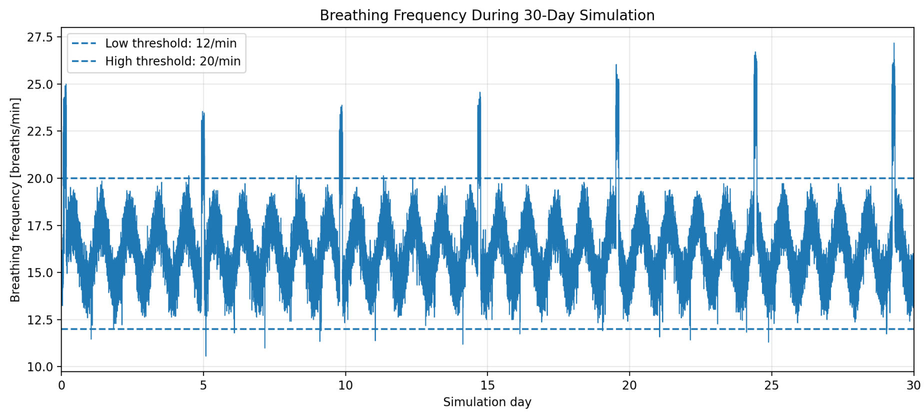

Figure 2: Breathing frequency across the 30-day simulation, with Low/High thresholds marked.

episodes are the clinically meaningful signal here: because they last 10 minutes or more, they are unlikely to be noise, similar to how real respiratory monitors use duration-based alerts (Hieronymi et al., 2011).

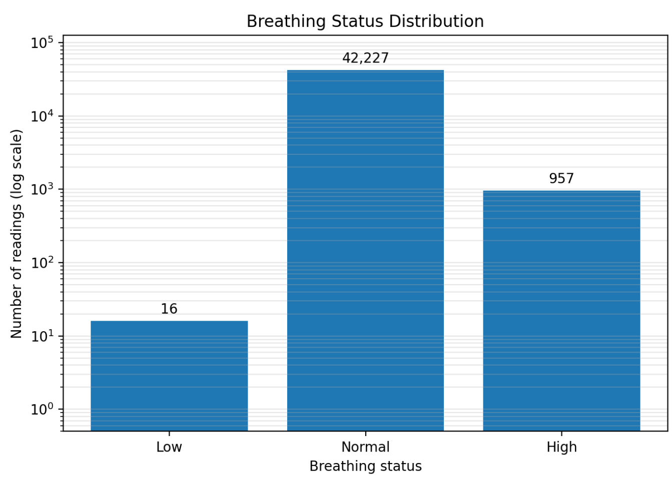

Figure 3: Distribution of breathing-status classes (log scale, with counts labelled).

5 Discussion

Strengths: the estimator never sees the hidden true breathing rate, only the raw signal, so the results are a genuine test of the detection method. All scripts share one data folder and one source of logic, so the notebook and the scripts cannot drift apart.

Limitations: the signals do not model motion artifacts or sensor noise from movement. The peak-detection thresholds are fixed, not adaptive like in Pan–Tompkins (Pan and Tompkins, 1985). The dashboard is a local desktop app, not a connected phone app like real products (Zheng et al., 2017).

Possible improvements: adaptive thresholds for both peak detectors, some simulated motion noise for a harder test, and a small automated test that checks a known input gives the expected rate.

6 Conclusion

This project simulates a full chestband pipeline: signal generation, rate estimation from a raw signal (not a stored value), threshold classification, and a live readout. The thresholds and detection method match established theory and real chest-strap products (Hieronymi et al., 2011; Cretikos et al., 2008; Zheng et al., 2017; Avila and Sampogna, 2011). Over 30 simulated days, the system correctly found 17 sustained High episodes as the meaningful signal in an otherwise normal dataset. A waveform bug found during testing was fixed by reusing the same buffer technique already used correctly elsewhere in the code.

7 Declaration of AI Tool Usage

This project used Claude (Anthropic) as a technical partner for:

- **Debugging:** finding the cause of the waveform-continuity bug and a file-path mismatch between scripts.
- **Code cleanup:** renaming and organising the project into one consistent file structure.
- **Literature support:** finding and checking sources (the Springer Handbook of Medical Technology, the Pan–Tompkins paper, and related chest-strap literature) for design choices already made in the code.
- **Report formatting:** converting the project's results and reasoning into this LaTeX report.

All AI suggestions were reviewed and adapted by the author. The system design, thresholds, and simulation logic are the author's own work.

References

Avila, N. and Sampogna, C. (2011). e-health – ambient assisted living and personal health systems. In Kramme, R., Hoffmann, K.-P., and Pozos, R. S., editors, *Springer Handbook of Medical Technology*, pages 1217–1240. Springer.

Cretikos, M. A., Bellomo, R., Hillman, K., Chen, J., Finfer, S., and Flabouris, A. (2008). Respiratory rate: The neglected vital sign. *Medical Journal of Australia*, 188(11):657–659.

Hieronymi, U., Kramme, R., and Kronberg, H. (2011). Respiratory monitoring and pulse oximetry. In Kramme, R., Hoffmann, K.-P., and Pozos, R. S., editors, *Springer Handbook of Medical Technology*, pages 971–986. Springer.

Kramme, R. (2011). Basic diagnostics in cardiology. In Kramme, R., Hoffmann, K.-P., and Pozos, R. S., editors, *Springer Handbook of Medical Technology*, pages 65–100. Springer.

Pan, J. and Tompkins, W. J. (1985). A real-time qrs detection algorithm. *IEEE Transactions on Biomedical Engineering*, BME-32(3):230–236.

Zheng, J., Ha, C., and Zhang, Z. (2017). Design and evaluation of a ubiquitous chest-worn cardiopulmonary monitoring system for healthcare application: A pilot study. *Medical & Biological Engineering & Computing*, 55:283–294.